

Case Study - Police Management System

1. Requirements

1. The system shall allow authorized users to log in using a username and password.
2. The system shall allow police officers to create incident reports for reported crimes.
3. The system shall allow detectives to assign suspects and evidence to a case.
4. The system shall allow administrators to manage users and roles in the system.
5. The system shall allow officers to update the status of a case (Open, Under Investigation, Closed).
6. The system shall store all case-related data including suspects, evidence, and incident reports in a centralized database.
7. The system shall display a dashboard with statistics such as number of cases, officers, and recent incidents.
8. The system shall restrict access to sensitive information based on user roles.

2. Scenario 1

Creating an Incident Report

Initial assumption:

A police officer is logged into the system and has permission to create incident reports.

Normal:

A crime is reported to the police department. The police officer opens the Police Management System and selects the option to create a new incident report.

The officer enters details such as:

- location of the incident
- type of crime
- description of the event
- date and time

The system saves the incident report in the database and generates a new case.

What can go wrong:

The officer may enter incomplete or invalid information. In this case, the system displays an error message and asks the officer to complete the required fields.

Other activities:

While the officer is creating the report, other officers may be accessing or updating other cases in the system.

System state on completion:

The new incident report is saved and a case is created in the system.

3. Scenario 2

Assigning a Suspect to a Case

Initial assumption:

A detective is logged into the system and a case already exists.

Normal:

The detective opens the case details page and selects the option to add a suspect.

The detective enters the suspect's information such as:

- name
- age
- description
- related evidence

The system stores the suspect information and links it to the selected case.

What can go wrong:

The detective may try to assign a suspect to a case that does not exist. In this situation, the system shows an error message.

Other activities:

Other officers may be adding evidence or updating case information at the same time.

System state on completion:

The suspect is successfully linked to the case.

4. Scenario 3

System Error or Unauthorized Access

Initial assumption:

A user attempts to access the system.

Normal:

The system verifies the username and password entered by the user.

If the credentials are correct, the user is granted access to the system based on their role.

What can go wrong:

If the credentials are incorrect or the user does not have permission to access certain features, the system denies access and displays an error message.

Other activities:

Other authorized users may continue using the system normally.

System state on completion:

The system either grants access or denies the request depending on the authentication result.

Use Case	Actor	Description	Precondition	Postcondition
Login	All Users (Officer, Detective, Admin)	User logs into the system using username and password	User has valid credentials	User is authenticated and granted access according to role
Create Incident Report	Police Officer	Officer creates a new report for a crime	Officer is logged in and has permission	New incident report is saved and a case is created
Assign Suspect to Case	Detective	Detective links a suspect to an existing case	Case exists in the system	Suspect information is saved and linked to the case
Add Evidence	Detective/Officer	Officer or detective uploads evidence related to a case	Case exists in the system	Evidence is stored and associated with the correct case
Update Case Status	Officer	Officer updates the status of a case (Open, Under Investigation, Closed)	Case exists	Case status is updated in the database
Manage Users & Roles	Administrator	Admin adds, deletes, or updates users and their roles	Admin is logged in	User list and roles are updated
View Dashboard	All Users	User views statistics: number of cases, recent incidents, number of officers	User is logged in	Dashboard displays up-to-date statistics
Access Control / Authorization	All Users	System restricts access to sensitive features based on user role	User is logged in	Users can only access features allowed for their role